OFFICIAL USER MANUAL & OPERATING GUIDELINES

ErrinILS

School Integrated Library System

Installation, daily circulation desk, automated 5-source cataloguing, student management, collection weeding, QR remote device pairing, analytics, and administrative recovery.



Version 1.3.1 (Windows Edition)

Zero-Dependency Architecture • Standalone C# Executable

Target OS:	Windows 7, 8, 8.1, 10, 11 (32/64-bit)	**Runtime:**	Microsoft .NET Framework 4.0+
Database:	Single JSON File (`errin-data.json`)	**Release Date:**	November 2026
License:	MIT Open Source License	**Web Portal:**	https://rzsy.github.io/ErrinILS/

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Quick Role Guidance

Librarians & Administrators: Read all chapters. Start with Chapters 3 to 5, then proceed to Chapters 6 through 16.
Student Helpers & Prefects: Focus on Chapter 4.4 (scanners), Chapter 7 (Circulation Desk), and Chapter 8 (Simplified Kiosk).
IT Technicians: Focus on Chapters 2, 3, 13 (Remote QR), 15 (Backups), and 18 (Troubleshooting).

1. About ErrinILS

1.1 What ErrinILS is

ErrinILS is an integrated library system (ILS) designed specifically for school libraries. It is a lightweight Windows desktop program that maintains synchronized, audit-proof records of three operational elements:

- The **books and learning items** in the library collection.
- The **students, teachers, and patrons** registered to borrow them.
- The **active loans, historical borrowings, holds (reservations), and fines** connecting patrons and items.

Everyday library operation does not need an internet connection. All information is kept on the library computer and is saved automatically each time you make a change. The internet is used only for two optional tasks: looking up book bibliographic details by ISBN, and searching remote library catalogues via Z39.50.

1.2 Core Philosophy & Zero-Dependency Architecture

Traditional library systems (such as Koha or SLiMS) require heavy server stacks: Apache, PHP, MySQL, and Linux administration. ErrinILS runs on a **zero-external-dependency rule**. Built with pure C# on base .NET Framework 4.0 assemblies (`mscorlib`, `System`, `System.Drawing`, `System.Windows.Forms`), it compiles into a standalone ~200 KB executable (`ErrinILS.exe`) storing all data in human-readable JSON (`%APPDATA%\Errin\errin-data.json`).

2. System Requirements & Hardware Guidelines

ErrinILS is deliberately light, running smoothly on older classroom PCs without servers or database daemons.

Component	Minimum Requirement	Recommended Specification
Operating System	Windows 7 SP1 (32 or 64-bit)	Windows 10 or Windows 11 (64-bit)
Runtime	Microsoft .NET Framework 4.0	Microsoft .NET Framework 4.8 (built-in on Win 10/11)
Memory (RAM)	1 GB RAM	2 GB RAM or more
Disk Space	10 MB for program; ~50 MB for data	100 MB free storage
Display Screen	1024 × 600 (min: 900 × 600)	1366 × 768 or 1920 × 1080 (High-DPI PerMonitorV2)
Scanner	Keyboard wedge scanner (USB)	Standard USB 1D laser or 2D image scanner
Printer	Optional	A4 printer or thermal label printer (Code 128 barcodes)

3. Installing, Updating and Starting ErrinILS

3.1 Installing with Setup program

Download `ErrinILS-Setup-1.3.1.exe`. Double-click it, click **Yes** at the UAC prompt, and if Windows SmartScreen appears, click **More info** then **Run anyway**. Follow the wizard and click **Finish**.

3.2 Running without installing (portable copy)

Extract `ErrinILS-Windows.zip` to any folder or USB flash drive and double-click `ErrinILS.exe`. It runs portably and keeps data in the same user profile directory.

3.3 Where your data is stored

Everything lives in `%APPDATA%\Errin\` (press Win+R and type `%APPDATA%\Errin`):

File Name	Purpose and Contents
`errin-data.json`	Main database: catalog, patrons, loans, holds, fines, and policy rules.
`errin-data.bak`	Automatic rollback copy: saved before every write. Recovers if sudden power loss occurs.
`errin-config.json`	Administrator password hash (PBKDF2/SHA-256), Z39.50 settings, and paired device tokens.

6. The Dashboard & Daily Overview

The Dashboard is the first page you see upon launching ErrinILS. It provides an immediate overview of active operations without requiring any manual data entry.

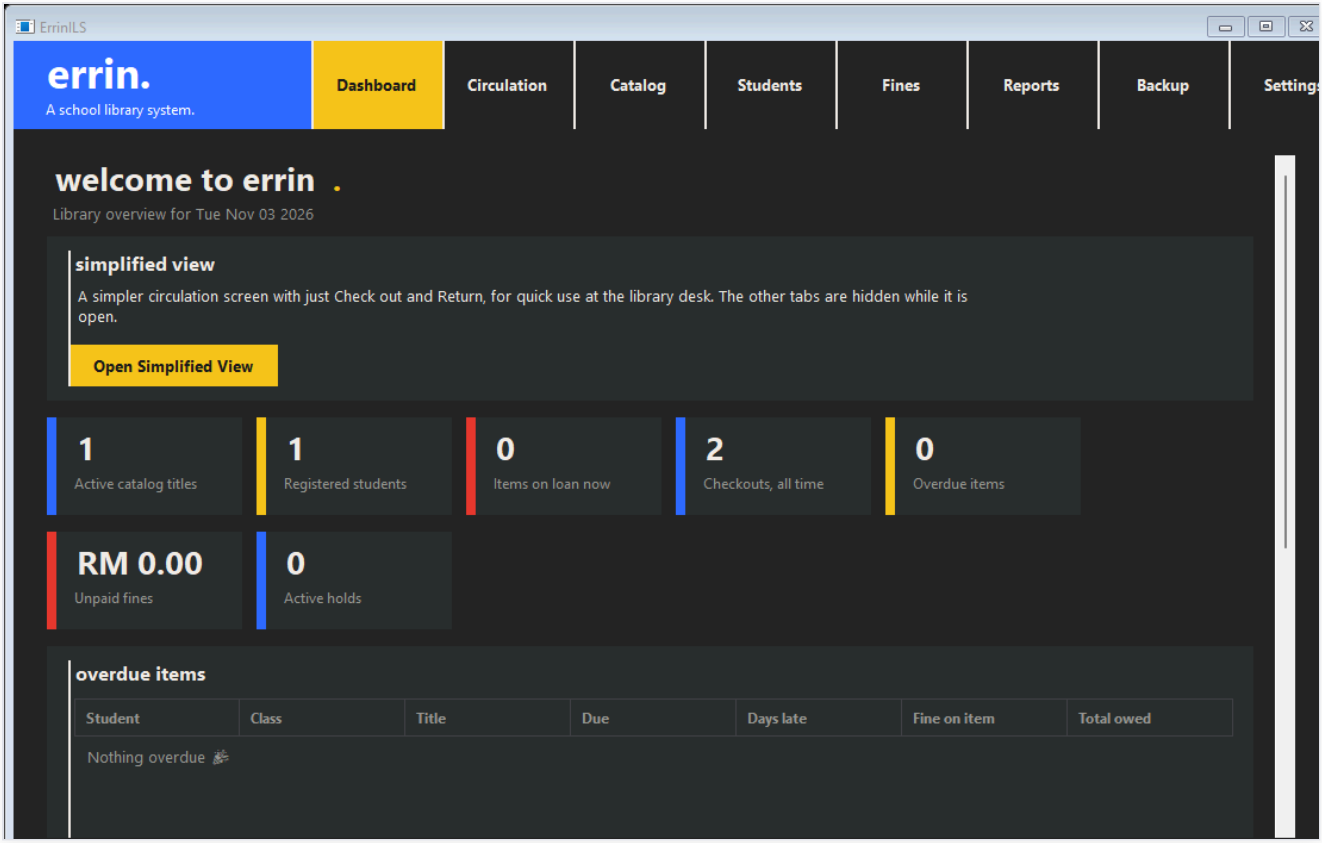


Figure 1: The ErrinILS Dashboard — Active collection metrics, real-time overdue notices table, and reading leaderboard.

6.1 What the Dashboard shows

- **Simplified View card:** One-click button to launch the student kiosk mode.
- **Metric Counters:** Active catalogue titles, registered students, books out on loan, all-time checkout tally, overdue book count, and total unpaid fines.
- **Overdue Items table:** Lists student name, class, book title, due date, days late, and total monetary balance owed. Use this table every morning to issue reminder slips.
- **Top 10 Borrowers:** Quick ranking of your top readers this month, this year, and of all time.

7. Circulation: Check Out, Return, Renew and Holds

Circulation is the busiest desk in the library. Click the **Circulation** tab; the cursor is automatically placed in the active scan box.

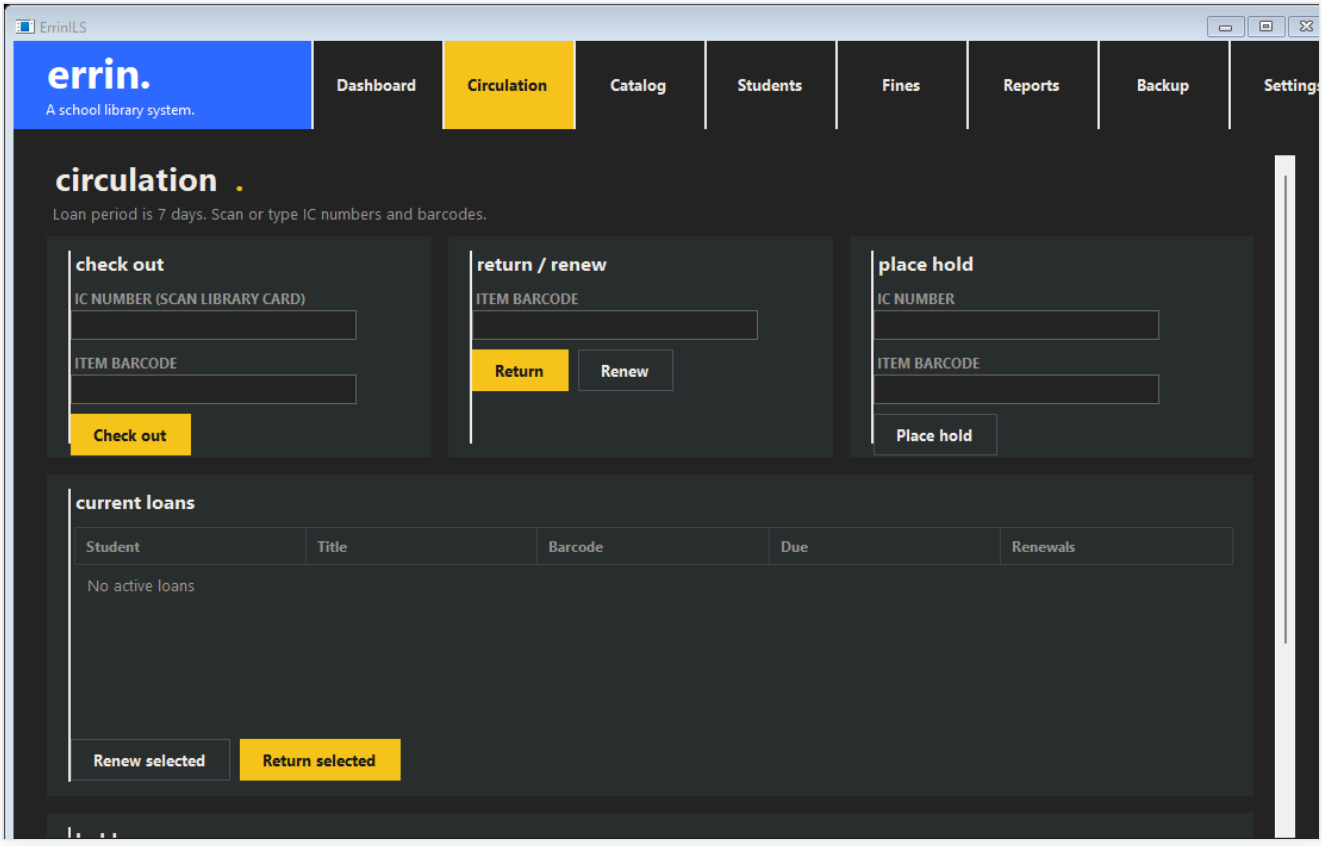


Figure 2: The Circulation Desk — Fast barcode checkout, returns/renewals box, active loans list, and holds queue.

7.1 Checking out a book

1. Scan the student's library card barcode into **IC number** (or type the 12 digits).
2. The cursor automatically advances to **Item barcode**. Scan the book's barcode.
3. The checkout registers immediately. Toast confirms: *"Checked out. Due [YYYY-MM-DD]"*.

7.2 Returning and renewing books

- **Return:** Scan the book barcode in **Return / Renew** and press Enter. If late, the fine is assessed immediately. If another student placed a hold, an alert notifies you to put the book aside.
- **Renew:** Scan the book and click **Renew**. Extends loan by 7 days. Blocked if overdue, renewed twice already, or on hold for another reader.

8. The Simplified View (Student Self-Checkout Kiosk)

A calm, high-contrast, large-print desk screen with only Check Out and Return, tailored for young students. Administrative toolbars are hidden.

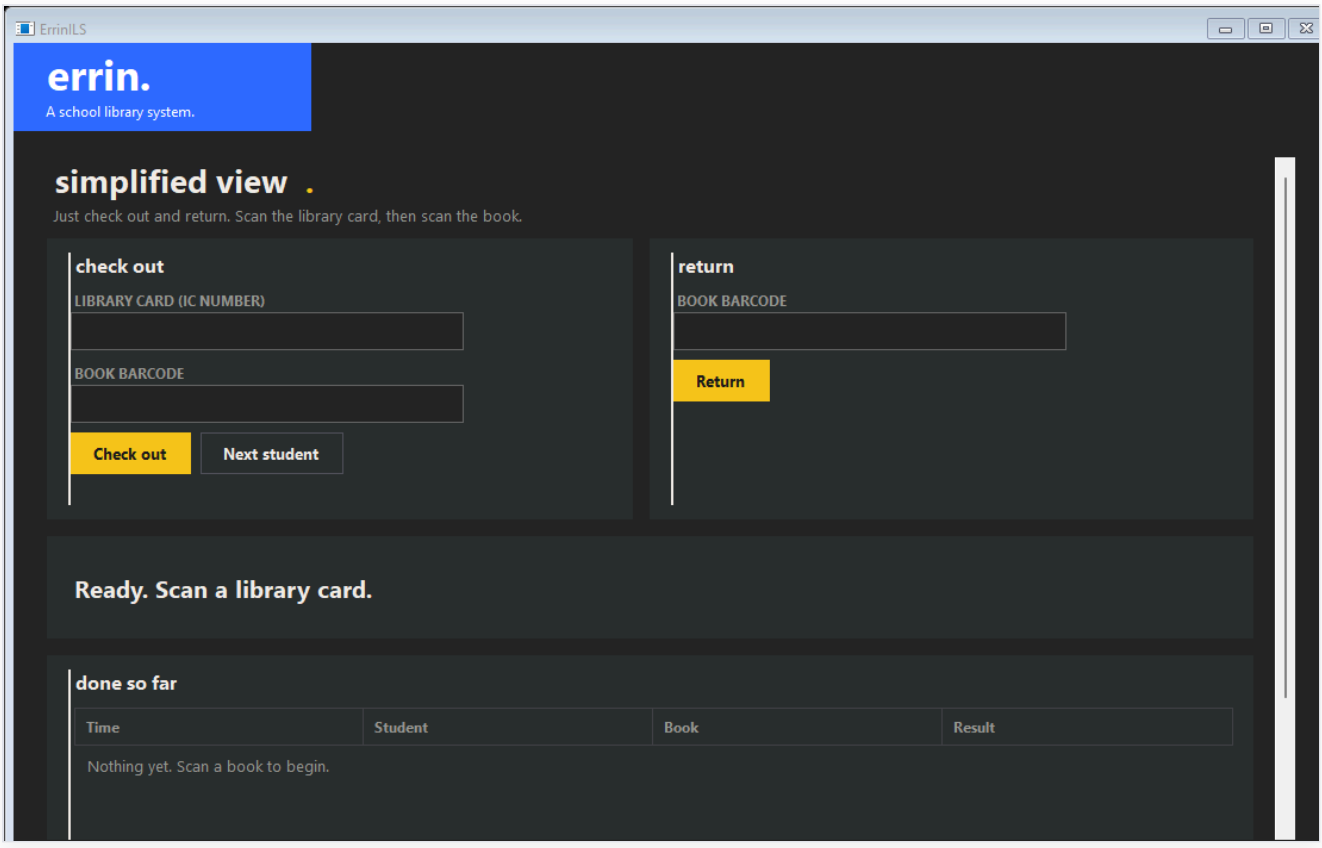


Figure 3: Simplified Student Kiosk — Child-friendly touch interface, large color banner, and audit transaction log.

8.1 How students use the kiosk

1. The screen prompts: *"Ready. Scan a library card."*
2. Student scans their card. The banner turns green: *"Hello, [Student Name]!"*
3. Student scans the book barcode. The kiosk plays a cheerful chime: *"Checked out: [Title]. Due back [Date]!"*
4. Click **Next student** to clear the screen for the next reader.

> **Exiting Kiosk Mode**
>
> Click **Exit Simplified View** in the bottom right corner. Entering the administrator password is required to return to the full administrative desk.

9. The Catalog & 5-Source ISBN Copy-Cataloguing

The Catalog tab manages all bibliographic records and physical copies.

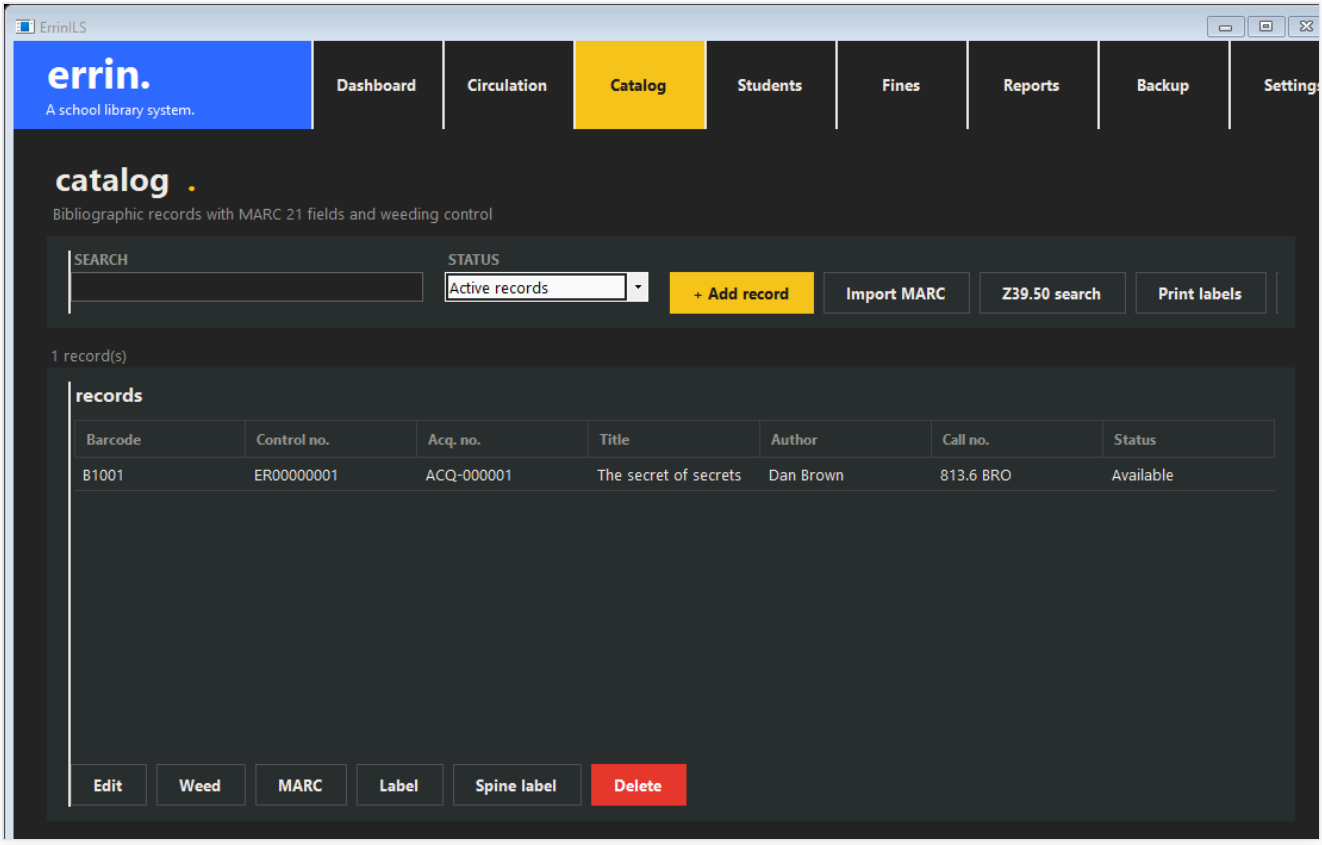


Figure 4: The Bibliographic Catalog — Multi-term searching, active vs weeded status filtering, and barcode generation.

9.1 5-Source ISBN Copy-Cataloguing

To add a book without manual typing, click **+ Add record**, scan the ISBN in **Copy cataloguing**, and click **Look up**. ErrinILS queries 5 international sources simultaneously:

1. **Library of Congress (Z39.50)**: Direct ASN.1 BER connection (`lx2.loc.gov:210`).
2. **Open Library**: REST API bibliographic repository.
3. **Google Books API**: Commercial book metadata, summaries, and publication years.
4. **HathiTrust**: Academic union catalogue via MARCXML.
5. **K10plus German Union**: SRU MARCXML gateway.

Title, author, publisher, publication year, subjects, and Dewey Decimal classification are auto-populated.

10. Students & Patron Management

Manage registered students, track class allocations, and print scannable library cards.

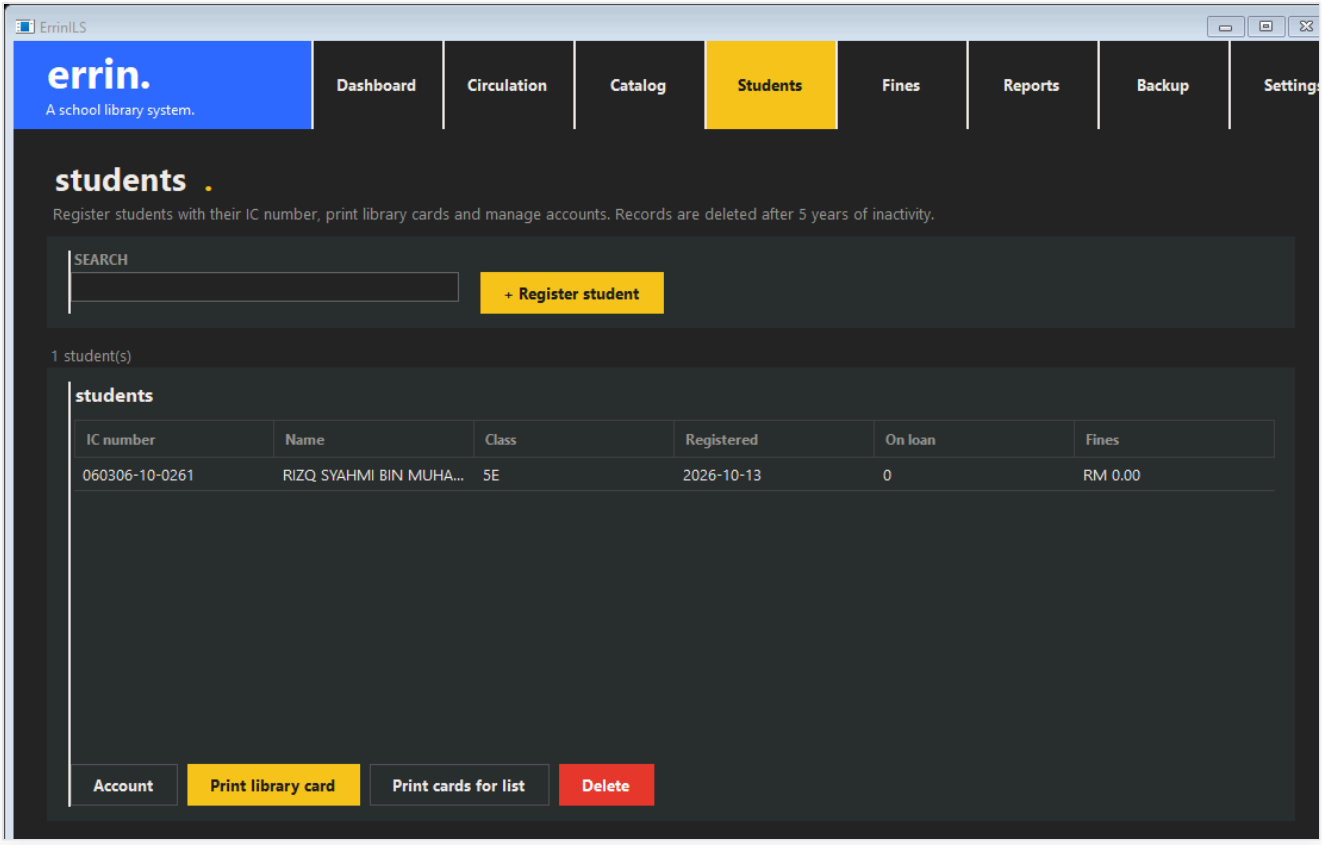


Figure 5: Student & Patron Management — Malaysian IC verification, borrowing histories, and card printing.

10.1 Registering students & Malaysian IC verification

Click **+ Register student**. Enter the 12-digit Malaysian IC or MyKid number. ErrinILS formats the number into `YYMMDD-PB-###G` and strictly verifies that the birth month and calendar day are mathematically valid. Enter the student's full name, class (e.g. `4 Melur`), and print their scannable card (85.6 × 54 mm).

11. Fines Management & Lost Books

Fines are assessed when late books are returned: `Days Overdue × Daily Rate (RM 0.10)`.

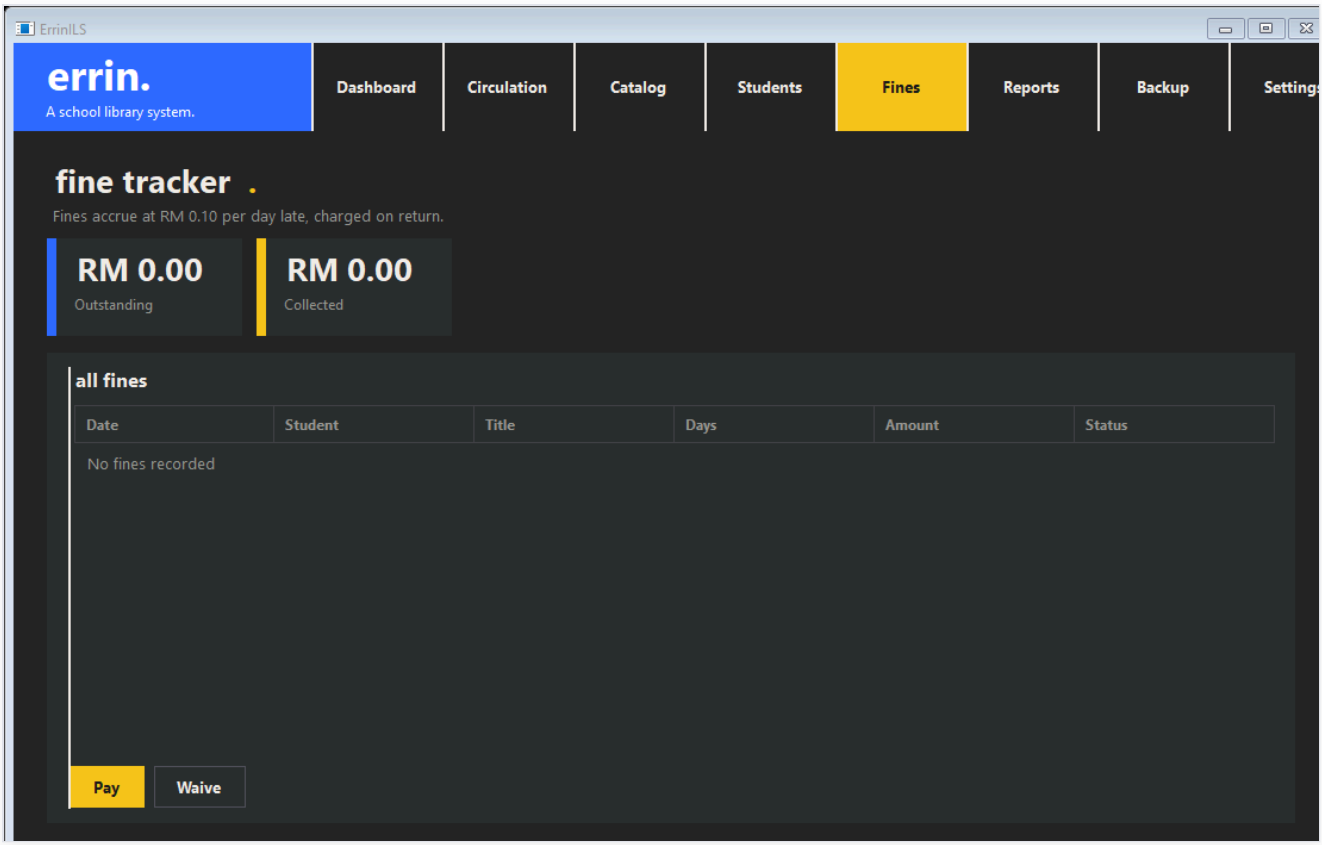


Figure 6: The Fine Tracker — Real-time outstanding balances, payment recording, and fine waivers.

Select any fine record in the table and click **Pay** when the student pays cash, or **Waive** if excusing the fee. Settling fines immediately lifts the borrowing block.

13. Remote Access, QR Code Pairing & Settings

ErrinILS features a built-in pure C# HTTP server (`0.0.0.0`) enabling wireless access from smartphones and tablets across school Wi-Fi and hotspots.

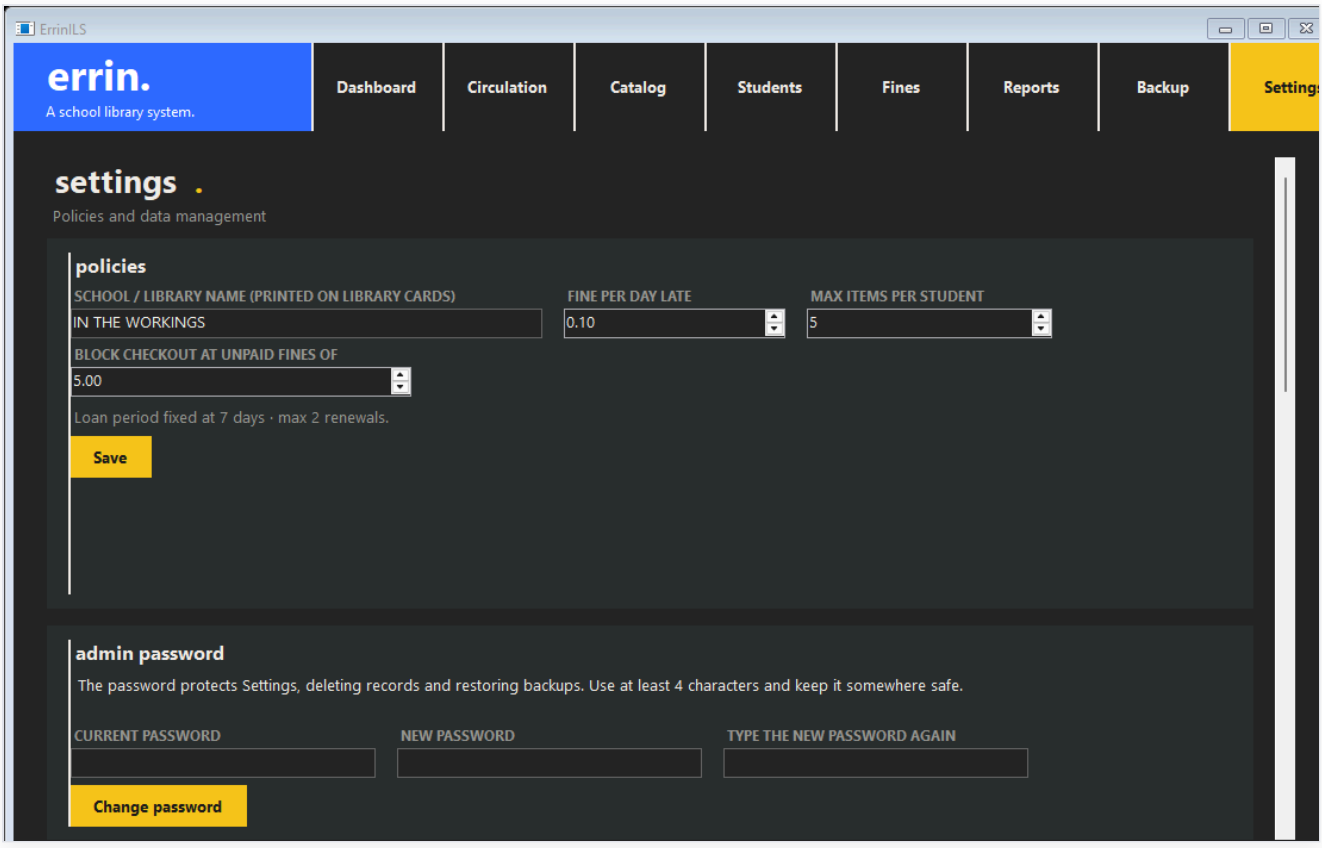


Figure 8: Settings & Remote Access — Fine rates, administrator password, and wireless QR code device pairing.

13.1 Instant QR Code & PIN pairing

1. In **Settings** → **Remote Access**, click **Pair New Device**.
2. Scan the generated QR code with your smartphone camera.
3. The mobile browser opens the web app with the 6-digit PIN pre-filled. Tap **Verify & Pair**.
4. A persistent 32-byte cryptographic token (`X-Errin-Token`) authenticates the device. Use your phone camera as a wireless shelf scanner.

14. Reports, Reading Champions & Collection Analytics

Open the **Reports** tab for executive library intelligence and Nilam reading program tracking.

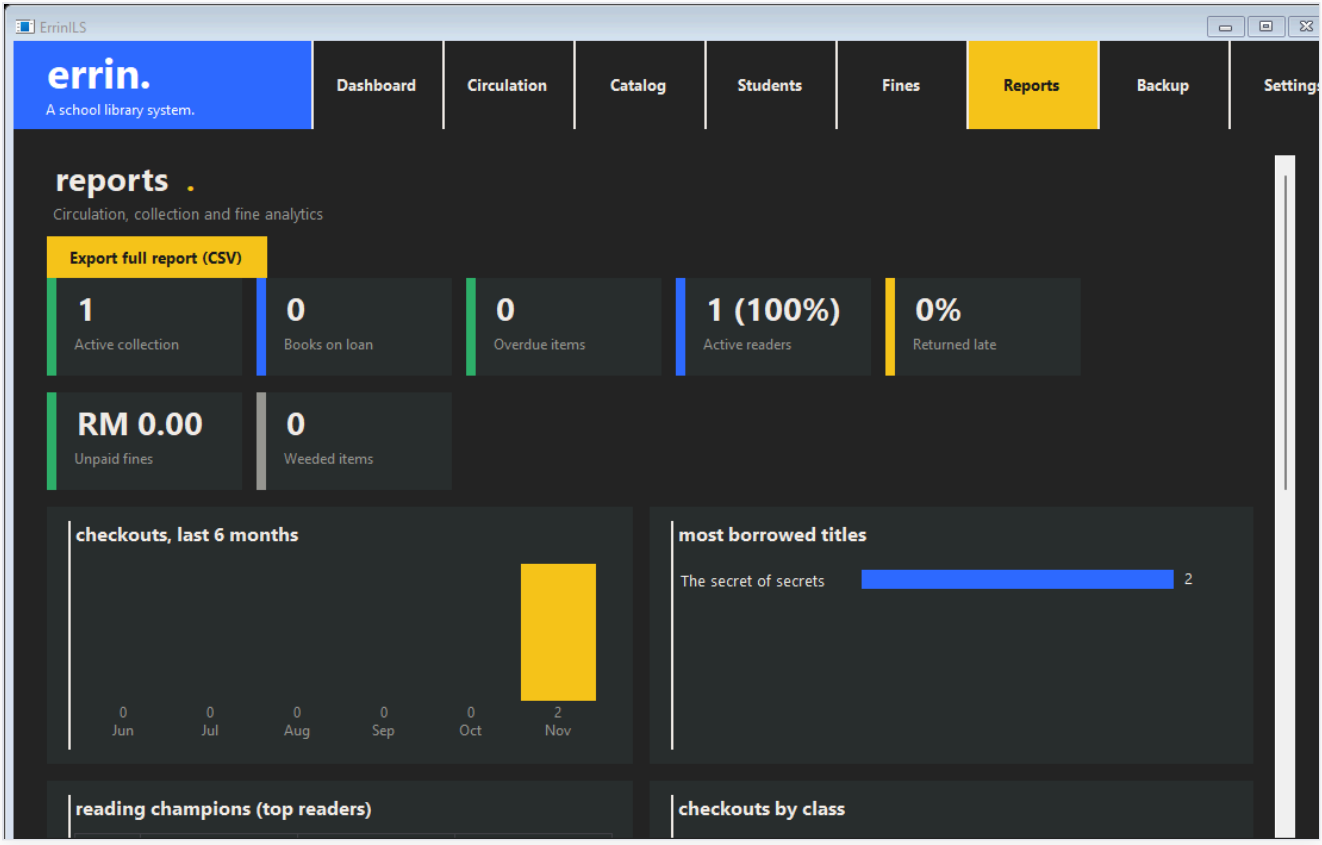


Figure 7: Reports & Reading Champions — Top 25 reader leaderboard, 6-month borrowing trajectory, and Dewey 100s breakdown.

- **Reading Champions Leaderboard:** Top 25 student readers ranked by total books borrowed.
- **Circulation Trajectory:** Monthly borrowing trends over the last 6 calendar months.
- **Dewey Decimal Breakdown:** Collection distribution across the 10 core DDC classes.
- **Dormant Inventory Audit:** Identifies titles with zero borrowings over the past 12 months.
- **1-Click CSV Full Report:** Exports a complete spreadsheet ready for school board presentations.

15. Backup & Disaster Recovery

In the **Backup** tab, click **Export Backup (ZIP)**. Save the file to an external USB flash drive every Friday afternoon. To restore on a new PC, install ErrinILS, open the Backup tab, and click **Restore from backup**.

18. Troubleshooting Guide

Problem Encountered	Likely Root Cause	Exact Solution
"Windows protected your PC"	Unsigned open-source installer.	Click *More info*, then click *Run anyway*.
Setup reports missing .NET	Windows 7 without .NET runtime.	Install free Microsoft .NET Framework 4.8.
Status bar red: "NOT saving"	Disk full or write-protected.	Free disk space; export emergency backup to USB.
Scanner does not submit	Scanner not sending Enter key.	Configure scanner in USB HID Keyboard mode with Enter suffix.
Phone cannot connect to QR Web	Phone on different Wi-Fi network.	Connect phone to the same Wi-Fi SSID as the PC.

Forgot Administrator Password?

Close ErrinILS, press `Win + R`, enter `%APPDATA%\Errin`, and rename or delete `errin-config.json`. Launch ErrinILS; the password resets to default (`admin` or `errin`). Set a new password immediately in Settings.

Appendix A. Quick Task Reference Guide

Goal	Tab	Action
Lend a book	Circulation	Scan student card, scan book barcode.
Return a book	Circulation	Scan book barcode in Return box.
Extend loan duration	Circulation	Scan book barcode, click Renew.
Add book via ISBN	Catalog	Click + Add record, scan ISBN, click Look up, Save.
Weed damaged book	Catalog	Select book, click Weed Copy, choose reason, confirm.
Register student	Students	Click + Register student, enter 12-digit IC, name, class.
Record fine payment	Fines	Select fine record, click Pay.
Pair phone via QR	Settings	Click Pair New Device, scan QR with camera.
Export backup to USB	Backup	Click Export Backup (ZIP), save to flash drive.

Appendix B. File and Directory Reference

Component	Standard File System Location
Program Binary	`C:\Program Files (x86)\ErrinILS\ErrinILS.exe`
Main Database	`%APPDATA%\Errin\errin-data.json`
Automatic Rollback File	`%APPDATA%\Errin\errin-data.bak`
Config & Auth Tokens	`%APPDATA%\Errin\errin-config.json`
Online Documentation	https://rzsy.github.io/ErrinILS/guide.html
Official Repository	https://github.com/RZSY/ErrinILS

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